

Sessional: $5+10+10+5 = 30$

MST: 25

EST: 45

6 questions, answer any 5..... (9*5)

Topics:

- a. General concepts about dockers, building, using them
- b. Running object detection, segmentation , classification on jetson
- c. Deepstream (source sink settings, config file for model)
- d. ROS (concepts of publisher/subscriber camera node, processor node, display node)
- e. Kubernetes (config files for pod/deployment/service/scaling)
- f. SLAM (linear/nonlinear formulation)
- g. Depth estimation (monocular, stereo camera model)

Question patterns:-

1. Application :

You need to mention which docker to use, type of problem (classification, detection etc), how will you approach it. The application may require depth concepts too.

2. Common issues with dockers

You need to remember the docker running, building , commit concepts and how they work

3. You may be given a sample situation:- ex I need to build an application which can be accessed by 10 users simultaneously, each user cannot use more than xxx resource etc
You have to demonstrate the use of Kubernetes, deployment/service whichever is required for your scenario
4. You may be given a robot navigation usecase and asked to model it.
You need to apply Kalman filter concepts, write the state matrix(F), control matrix etc and formulate it based on whether the model is linear (Kalman filter) or nonlinear (extended Kalman filter)
5. You can be given an application running mode: ex. I want to run a detection application using my camera and generate output as a video file with best possible speed.
Any mention of steaming applications refer to the use of deepstream, you need to mention the configurations you will use (source sink settings, model config etc)
The model config part overlaps with your MST concepts where you converted a model, wrote the config file.
6. When you need to combine robot movement with vision for any application, use ROS concepts. You should be familiar with setting it up (camera nodes, processor nodes, display nodes) , how you need to launch

Questions may require you to write the approach with setting up details and code snippets. Alternatively, you may get snapshots of deployed

application code and asked where you will need to change or identify the problem.